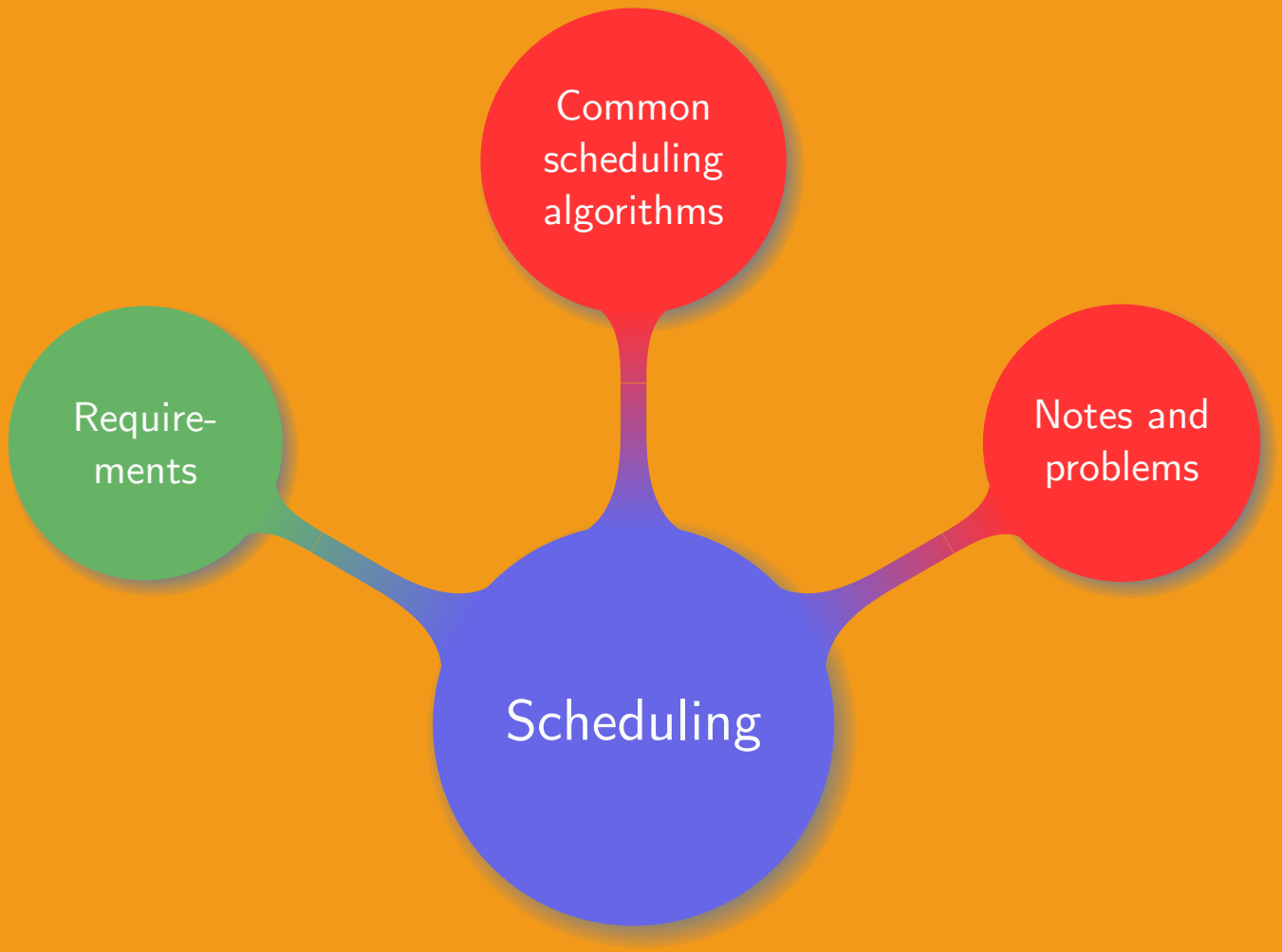
# Introduction to Operating Systems

## 4. Scheduling

Manuel – Fall 2019

# Chapter organisation



# Initial problem

Scheduler's job:

- Multiple processes competing for using the CPU
- More than one process in ready state
- Which one to select next?
- Key issue in terms of "perceived performance"
- Need "clever" and efficient scheduling algorithms

# Scheduler's job

Switching process is expensive:

- Switch from user mode to kernel mode
- Save state of current process (save register, memory map...)
- Run scheduling algorithm to select a new process
- Remap the memory address for the new process (convert address generated by the program into physical memory address in RAM); done by the Memory Management Unit (MMU)
- Start new process

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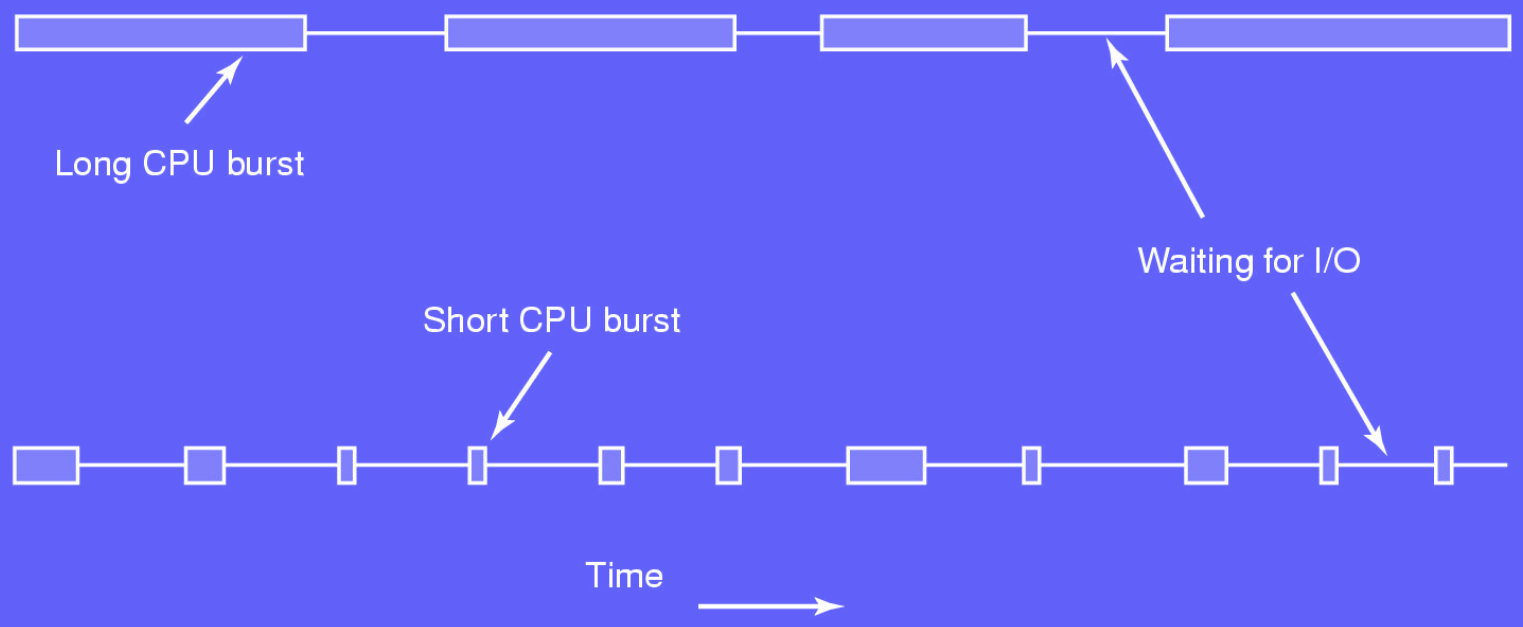
*Too many switches per second wastes much CPU*

# Process behavior




Typical behavior:

- Process runs for a while
- System call emitted to read/write from/in a file
- More general: process in blocked state until external device has completed its work

# Process behavior

Compute bound vs. I/O bound:

- Most time spent computing vs. waiting for I/O
- Length of the CPU burst:
  - I/O time is constant
  - Processing data is not constant
- As CPUs get faster processes are more and more I/O bound

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**Conclusion:** I/O bound processes should be run quickly, such as to issue their I/O request and keep the disk busy

# When to schedule?

Basic scheduling issues:

- Creation of a new process: both the parent and the child are in ready state
- A process exits: which process to run next? i
- A process blocks (semaphore, I/O...), what process to run?
- I/O interrupt from a device that has completed its task: run the newly ready process, or another one?

# Preemptive vs. non-preemptive

Two main strategies for scheduling algorithms:

1. Preemptive: a process is run for at most $n$ ms; if it is not done at the end of the period then it is suspended and another process is selected and run: require a clock interrupt

2. Non-preemptive: a process runs until it blocks or voluntarily releases the CPU; it is resumed after an interrupt unless another process with higher priority is in the queue

# What is the goal?

All systems:

- Fairness: fair share of the CPU for each process
- Policy enforcement: following the defined policy
- Balance: all parts of the system are busy

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Batch systems:

- Throughput: maximise the number of jobs per hour
- Turnaround time: minimise the time between submission and termination of a job
- CPU utilisation: keep the CPU as busy as possible

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Interactive systems:

- Response time: quickly respond to requests
- Proportionality: meet user's expectations

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Real-time systems:

- Meeting deadlines: avoid losing data
- Predictability: avoid quality degradation for multimedia

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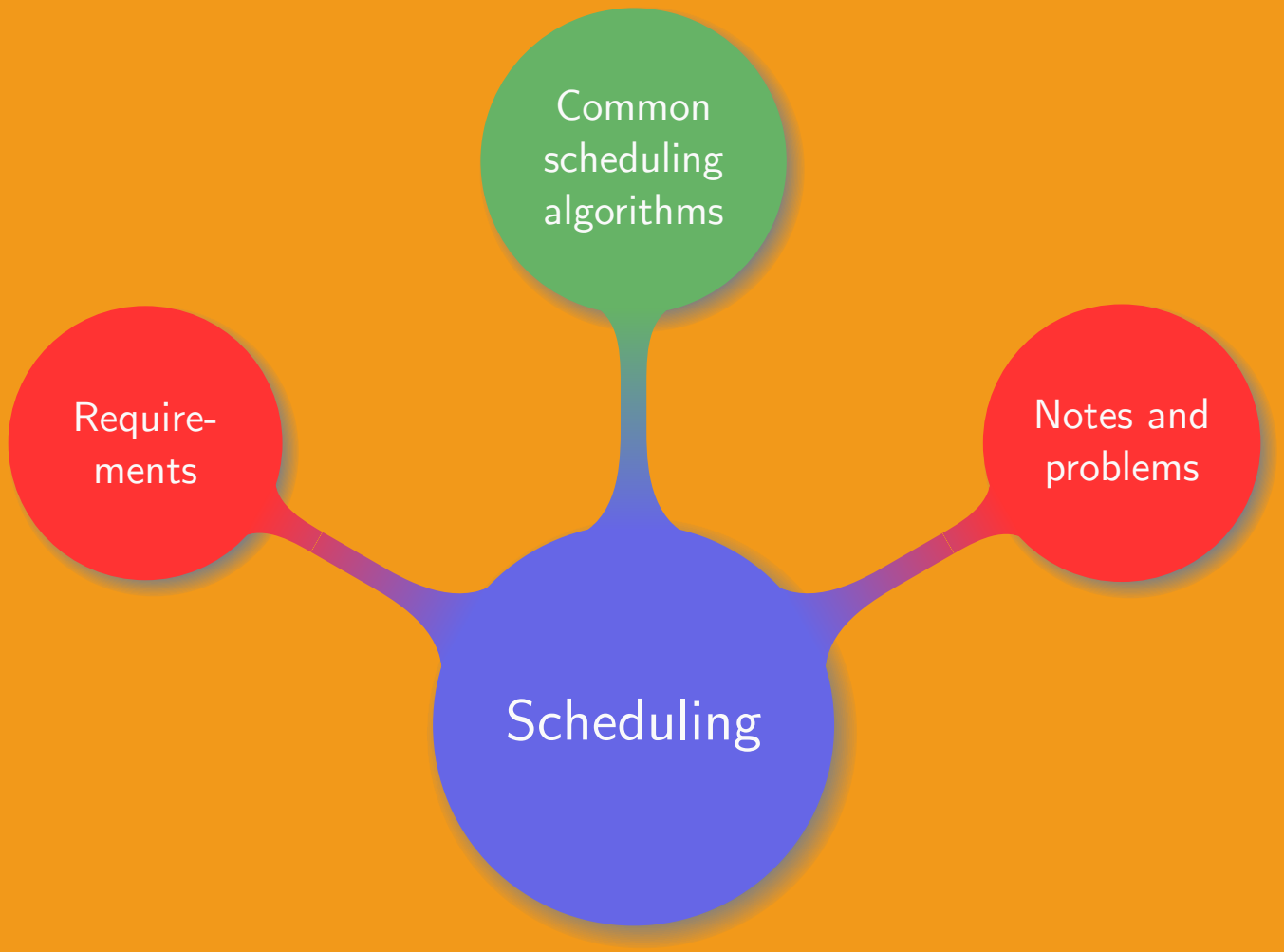

# First-come, first-served

**Basics:** simplest algorithm, non-preemptive

- CPU is assigned in the order it is requested
- Processes are not interrupted, they can run a long as they want
- New jobs are put at the end of the queue
- When a process blocks the next in line is run
- When a blocked process becomes ready it is put at the end of the queue

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**Characteristics:** good for batch systems, implemented using a single linked list

**Drawback:** what happens when there is one compute-bound process and many I/O-bound ones?

# Shortest job first

**Basics:** Non-preemptive, run times are known in advance

- Run time: A: 8 min, B: 4 min, C: 4 min, D: 4 min
- Turnaround time: $\frac{8+12+16+20}{4} = 14$ min

- Run time: B: 4 min, C: 4 min, D: 4 min, A: 8 min
- Turnaround time: $\frac{4+8+12+20}{4} = 11$ min

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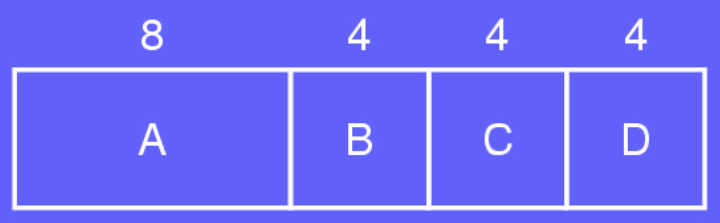


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- Run time: B: 4 min, C: 4 min, D: 4 min, A: 8 min
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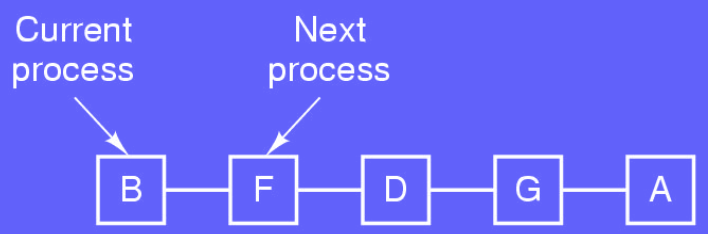
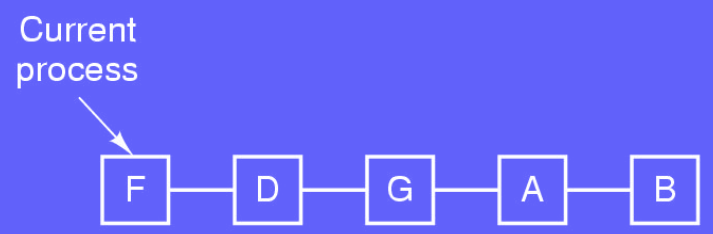
**Characteristics:** good for batch systems, very specific to a system

**Drawback:** what happens when the jobs are not all in a queue at the beginning?

# Round-Robin scheduling

**Basics:** preemptive, old, simple, fair, most widely used




- Each process is assigned a time interval called *quantum*
- A process runs until (i) it blocks, (ii) it is finished or (iii) its quantum is over
- Switching process is then done

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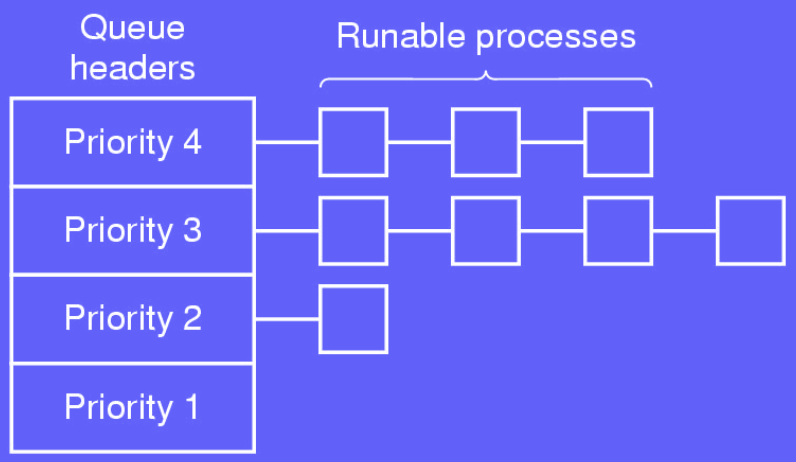
- Each process is assigned a time interval called *quantum*
- A process runs until (i) it blocks, (ii) it is finished or (iii) its quantum is over
- Switching process is then done

**Characteristics:** good for interactive systems, easy to implement (only needs to maintain a list of runnable processes)

**Drawback:** how long should a quantum be?

# Priority scheduling

**Basics:** preemptive, priority depending on who or what runs




- Processes are more or less important (e.g. printing...)
- Creates priority classes
- Use Round-Robin within a class
- Run higher priority processes first

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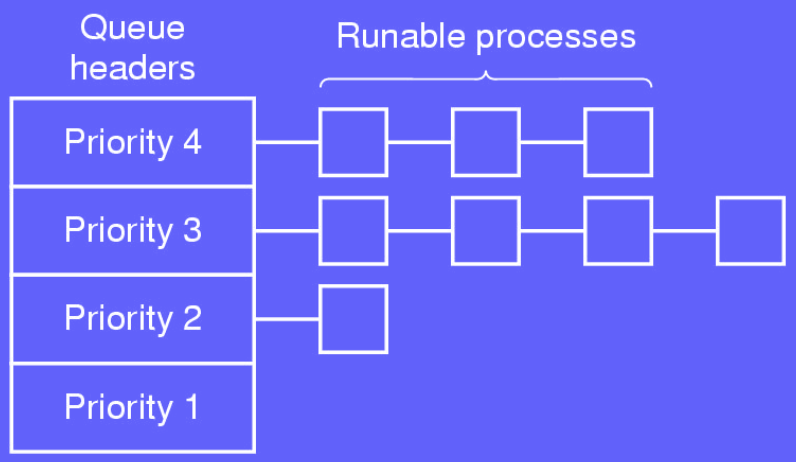


- Processes are more or less important (e.g. printing...)
- Creates priority classes
- Use Round-Robin within a class
- Run higher priority processes first

**Characteristics:** good for interactive systems, flexible, adjustable

**Drawback:** what happens if many high-priority processes run for a long time?

# Lottery scheduling

**Basics:** preemptive, extends priority scheduling

- Processes get "lottery tickets"
- When a scheduling decision is made a random ticket is chosen
- Price for the winner is to access resources
- High priority processes get more tickets (higher probability of winning)

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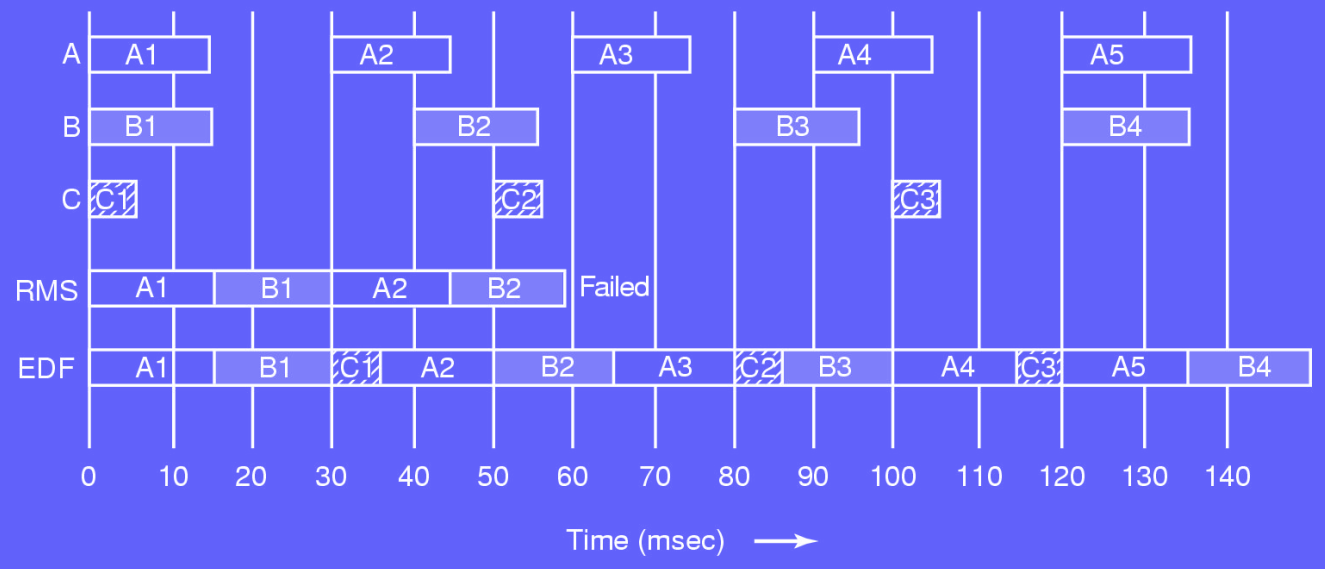
- Processes get "lottery tickets"
- When a scheduling decision is made a random ticket is chosen
- Price for the winner is to access resources
- High priority processes get more tickets (higher probability of winning)

**Characteristics:** good for interactive systems, highly responsive, possibility of cooperation between processes

**Drawback:** what if a "low level user" runs many small processes instead of a big one?

# Earliest deadline first

**Basics:** preemptive, priority based




- Process needs to announce (i) its presence and (ii) its deadline
- Scheduler orders processes with respect to their deadline
- First process in the list (earliest deadline) is run

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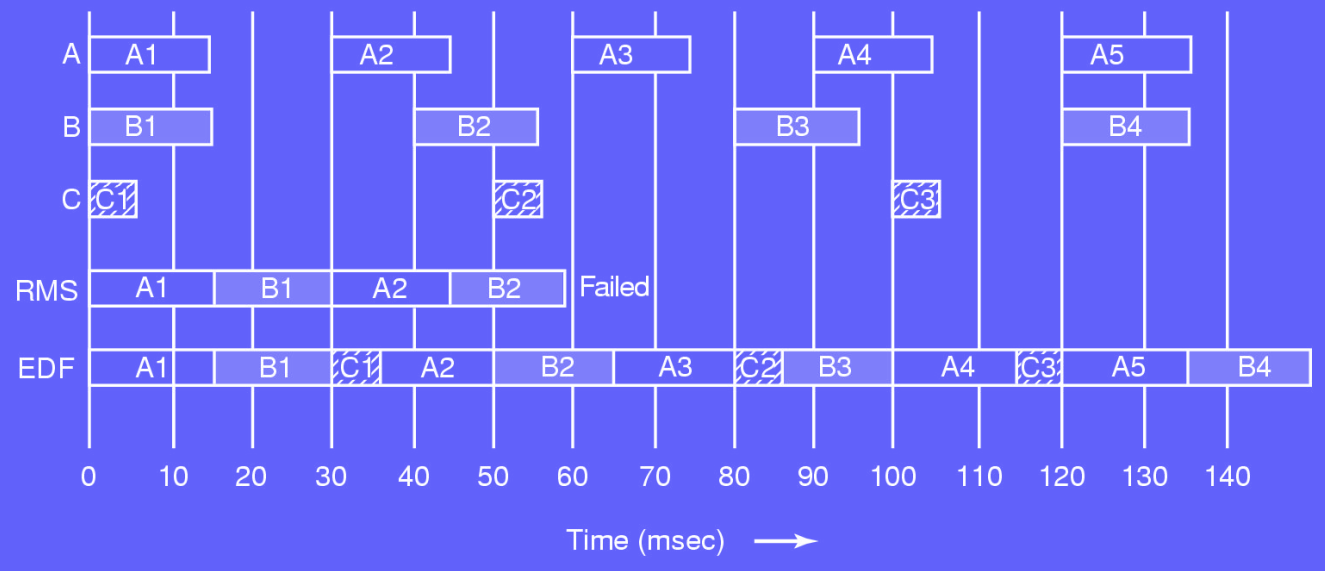
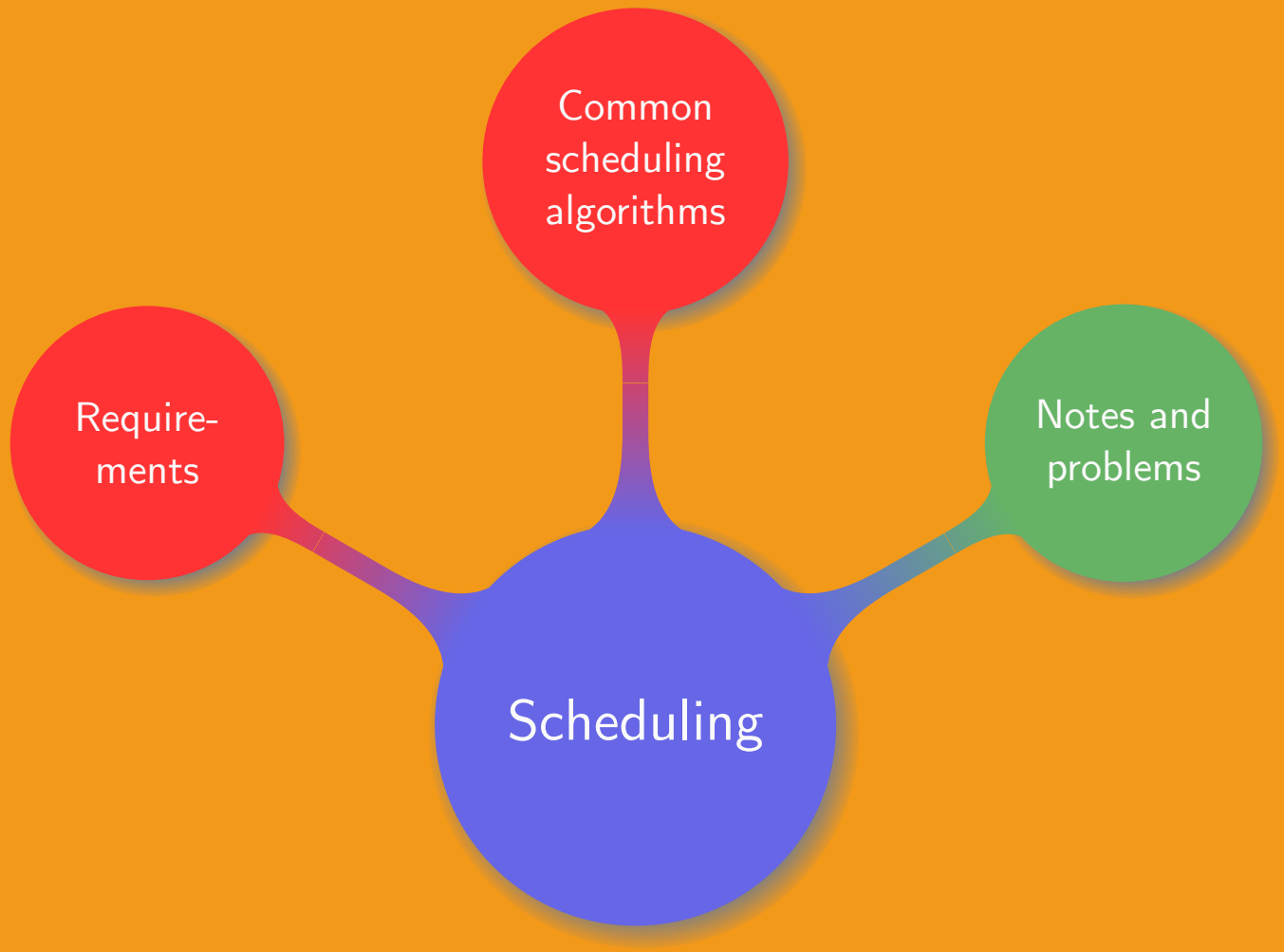



- Process needs to announce (i) its presence and (ii) its deadline
- Scheduler orders processes with respect to their deadline
- First process in the list (earliest deadline) is run

**Characteristics:** good for realtime systems, can fully use the CPU

**Drawback:** complex to implement

# Chapter organisation



# Policy vs. mechanism

Limitations of the previous algorithms:

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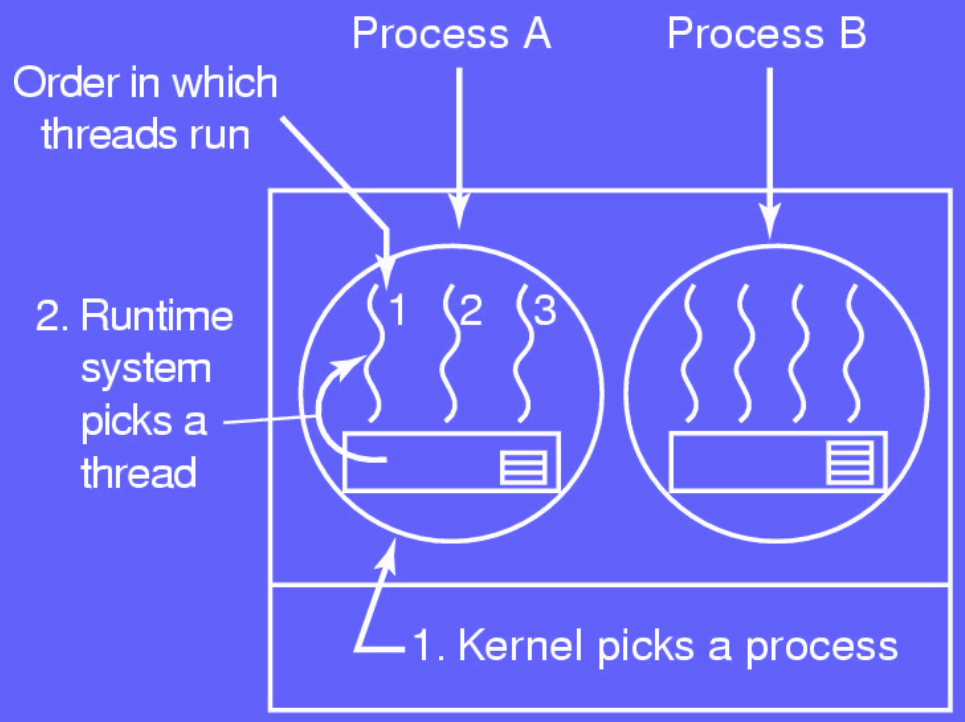
- Parent could know which of its children is most important

The solution consists in separating the scheduling mechanism from the scheduling policy:

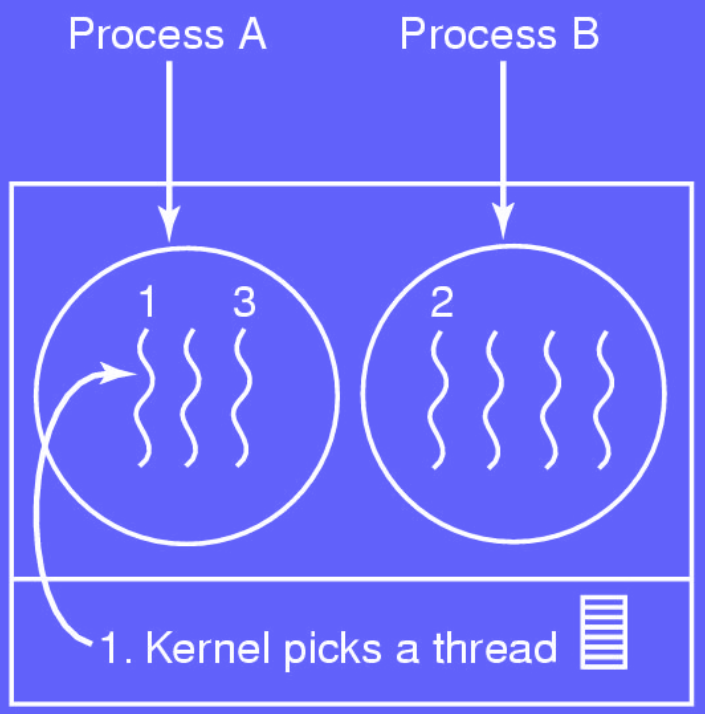
- Scheduling algorithm has parameters
- Parameters can be set by processes
- A parent can decide which of its children should have higher priority

# Threads scheduling – User-level




Possible: A1, A2, A3, A1, A2, A3
Impossible: A1, B1, A2, B2, A3, B3

# Threads scheduling – Kernel-level




Possible: A1, A2, A3, A1, A2, A3
Also possible: A1, B1, A2, B2, A3, B3

# The dining philosophers problem

Synchronisation problem:

- A philosopher is either thinking or eating
- When he is hungry he takes:
  1. His left chop-stick
  2. His right chop-stick
- Eats
- Puts down his chop-sticks
- Thinks

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First obvious solution:

- Wait for a chop-stick to be available
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**Problem:** all process start at the same time, then nobody ever eats

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- Philosopher thinks
- Lock mutex
- Acquire chop-sticks, eat, put down chop-sticks
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**Problem:** how many philosophers can eat at the same time?

# The dining philosophers problem

```
#define N 5
#define LEFT (i+N-1)%N
#define RIGHT (i+1)%N
enum { THINKING, HUNGRY, EATING };
int state[N]; mutex mut = 0 ; semaphore s[N];
void philosopher(int i) {while(TRUE) {think();take_cs(i);eat();put_cs(i);}}
void take_cs(int i) {
  mutex-lock(&mut);
  state[i] = HUNGRY; test(i);
  mutex-unlock(&mut); down(&s[i]);
}
void put_cs(int i) {
  mutex-lock(&mut);
  state[i] = THINKING; test(LEFT); test(RIGHT);
  mutex-unlock(&mut);
}
void test(int i) {
  if(state[i]==HUNGRY && state[LEFT]!=EATING && state[RIGHT]!=EATING;) {
    state[i]=EATING; up(&s[i]); }
}
```

# Key points

- Why is scheduling the lowest part of the OS?
- What are the two main types of algorithm?
- What are the two most common scheduling algorithms?
- Give an example of theoretical problem related to scheduling

Thank you!